



TARGETING THE UN-TARGETABLE : A Ferroptotic Approach to Refractory Cancers

BIOPC

Exploiting iron, lipid-peroxidation and antioxidant dependencies to identify vulnerabilities in therapy-resistant

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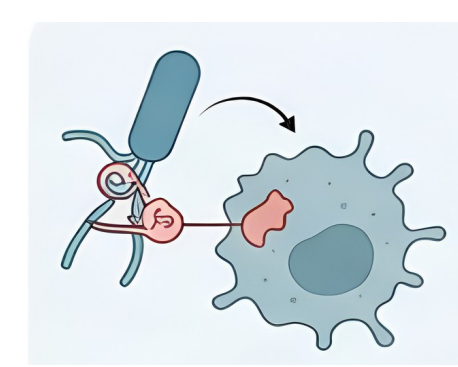
LITERATURE SYNTHESIS • PUBLIC-DATA ANALYSIS • COMPUTATIONAL HYPOTHESIS GENERATION

THE CENTRAL HYPOTHESIS : Refractory cancers may acquire or retain dependencies on iron metabolism, lipid remodeling and antioxidant defense. Ferroptosis exploits these vulnerabilities by overwhelming the cellular capacity to detoxify phospholipid peroxides.

THERAPEUTIC RESISTANCE → METABOLIC / REDOX ADAPTATION → IRON + PUFA-LIPID DEPENDENCY → ANTIOXIDANT DEFENSE BECOMES CRITICAL → FERROPTOTIC VULNERABILITY → POTENTIAL THERAPUTIC OPPORTUNITY

1. THE PROBLEM

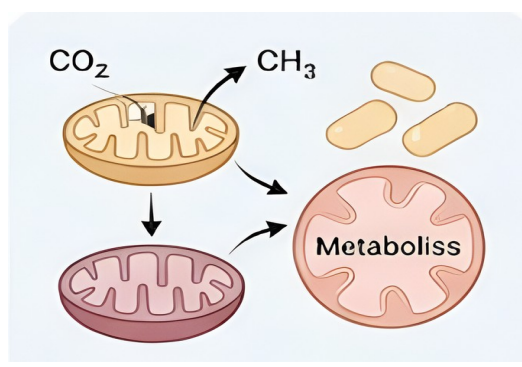
WHY DO REFRACTORY CANCERS REMAIN DIFFICULT TO TARGET?



01: TARGET ALTERATION

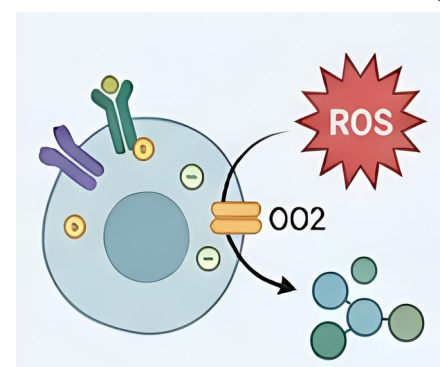
Mutations, pathway rewiring and loss of dependence on the original therapeutic target can reduce treatment efficacy.

Resistance to one mechanism may expose dependence on another survival system.



02: METABOLIC PLASTICITY

Resistant cells can remodel glucose, lipid, amino-acid and antioxidant metabolism to maintain survival under therapeutic stress.



03: REDOX ADAPTATION

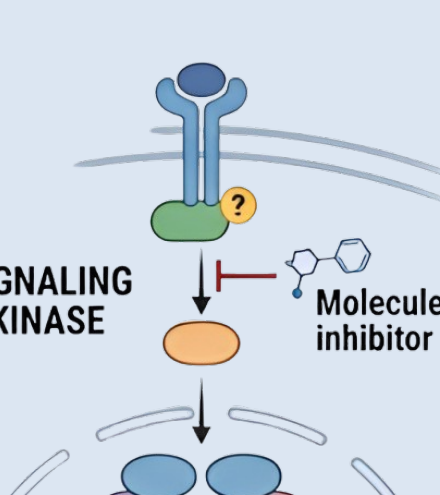
Enhanced ROS-buffering capacity can protect cancer cells from oxidative damage.

2. WHY FERROPTOSIS ?

FROM "UN-TARGETABLE" DRIVER → TARGETABLE VULNERABILITY

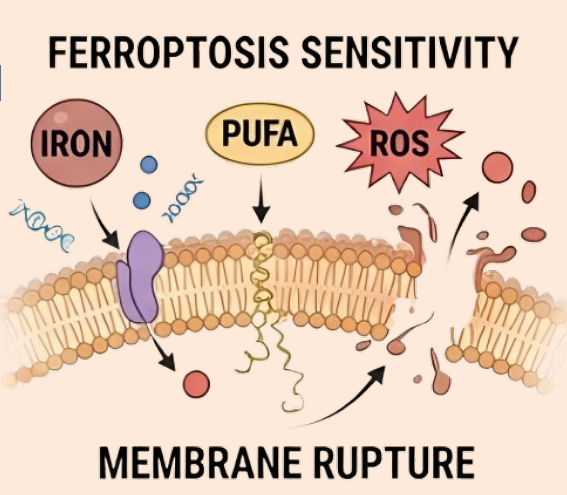
TRADITIONAL THERAPUTIC APPROACH

- Often centered on a specific driver
- Focuses on signaling
- Requires persistent target dependence



FERROPTOSIS VULNERABILITY FRAMEWORK

- Exploits iron/redox/lipid dependencies
- Focuses on membrane redox homeostasis
- Targets cellular metabolic state



3. LITERATURE EVIDENCE

Fe IRON

Iron availability contributes to oxidative pressure.

LIPIDS

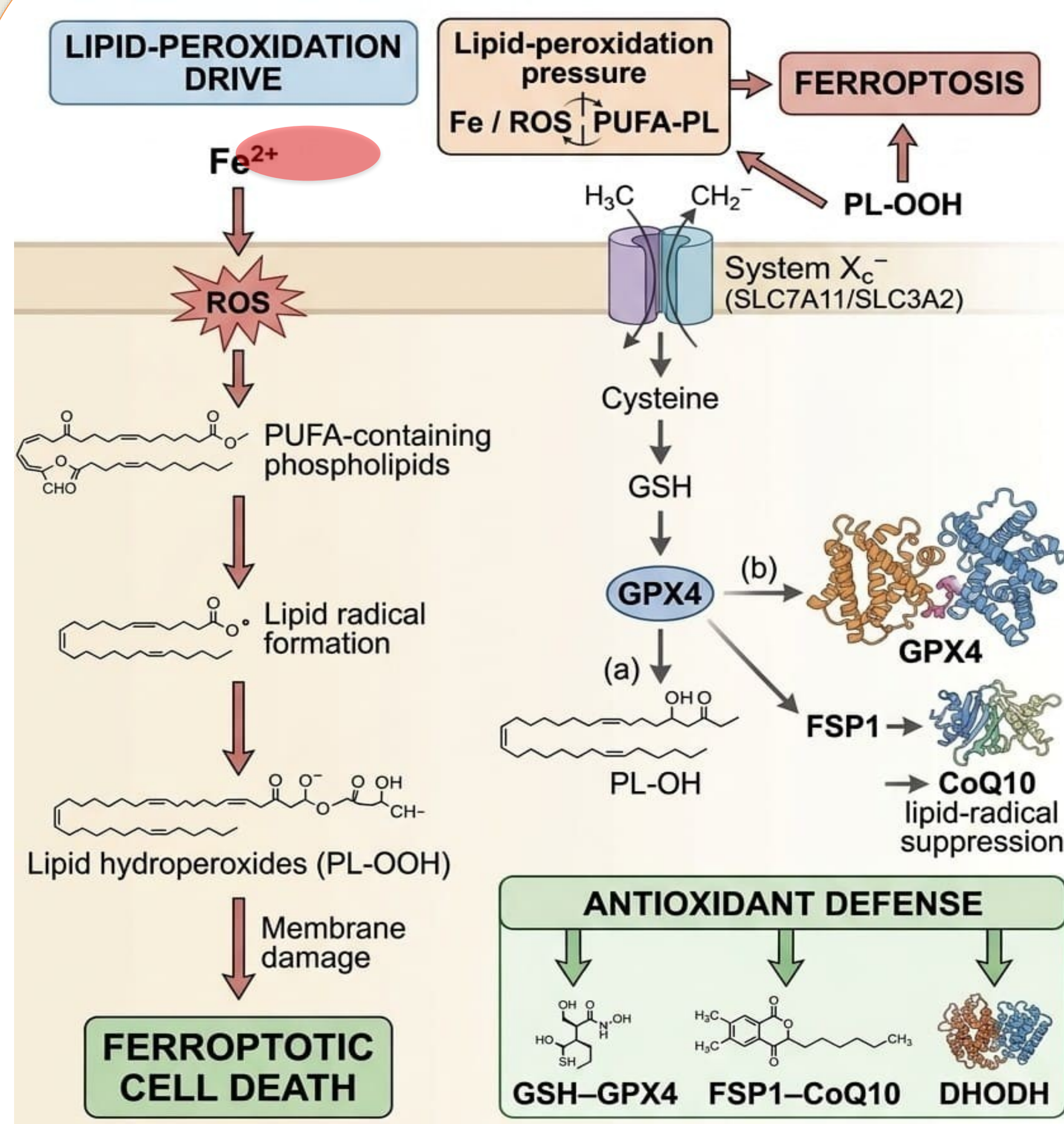
PUFA containing phospholipids influence peroxidation susceptibility..

ANTIOXIDANT DEFENSE

GPX4 and alternative defense pathways regulate ferroptotic sensitivity.

4. FERROPTOSIS BIOCHEMISTRY

THE BIOCHEMICAL ENGINE OF FERROPTOSIS

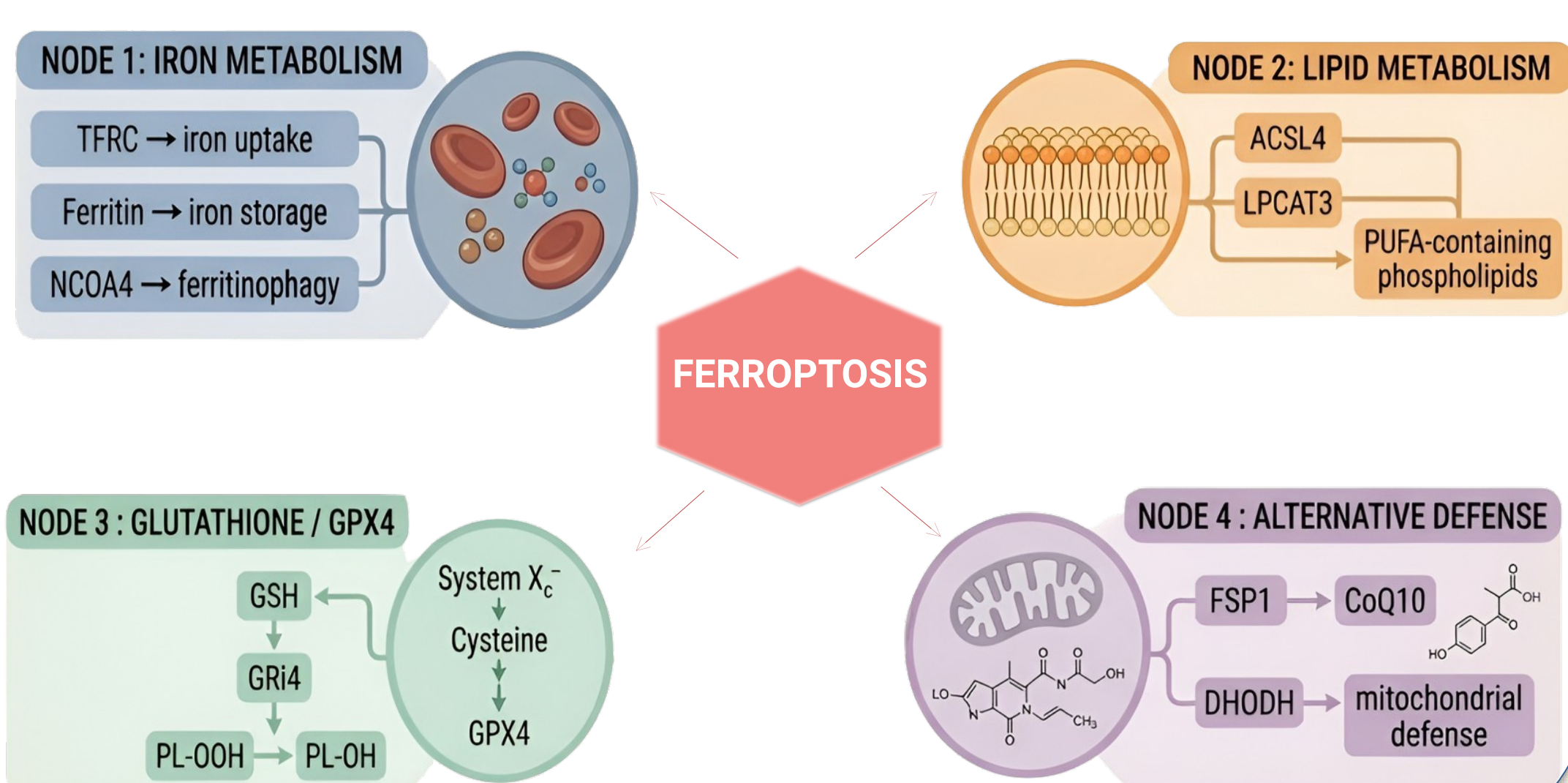


Ferroptotic susceptibility = Lipid peroxidation pressure balanced against antioxidant defenses.

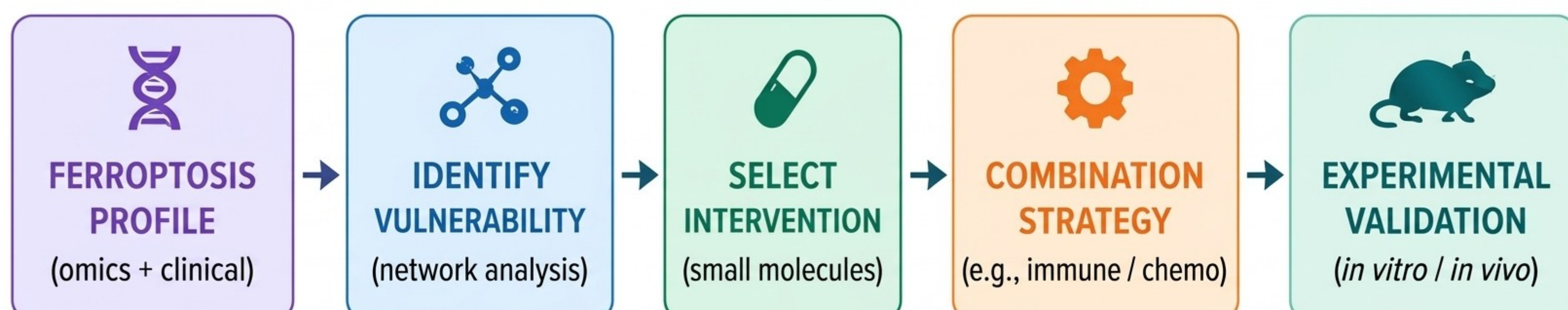
[Literature-derived mechanistic synthesis]

5. WHERE CAN WE TARGET ?

FERROPTOSIS AS A MULTI-NODE TARGETING NETWORK



8. PROPOSED THERAPEUTIC FRAMEWORK



Computational analysis generates candidates; experimental validation establishes causality.

9. CONCLUSION

THREE TAKE-HOME MESSAGES



RESISTANCE CAN CREATE DEPENDENCIES



FERROPTOSIS EXPLOITS REDOX-LIPID VULNERABILITY

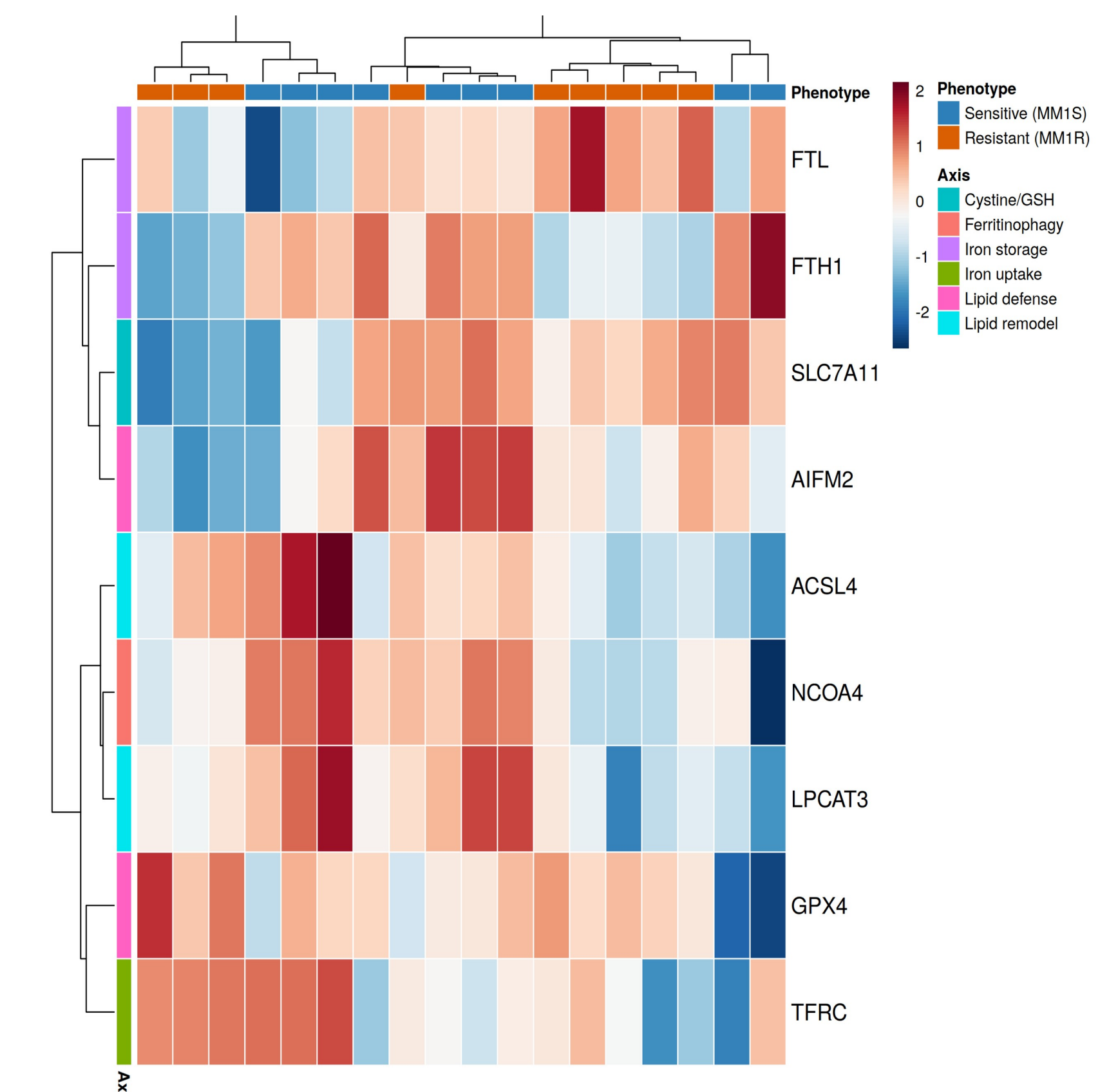


BIOMARKERS MAY ENABLE PRECISION TARGETING

6. PUBLIC DATA ANALYSIS

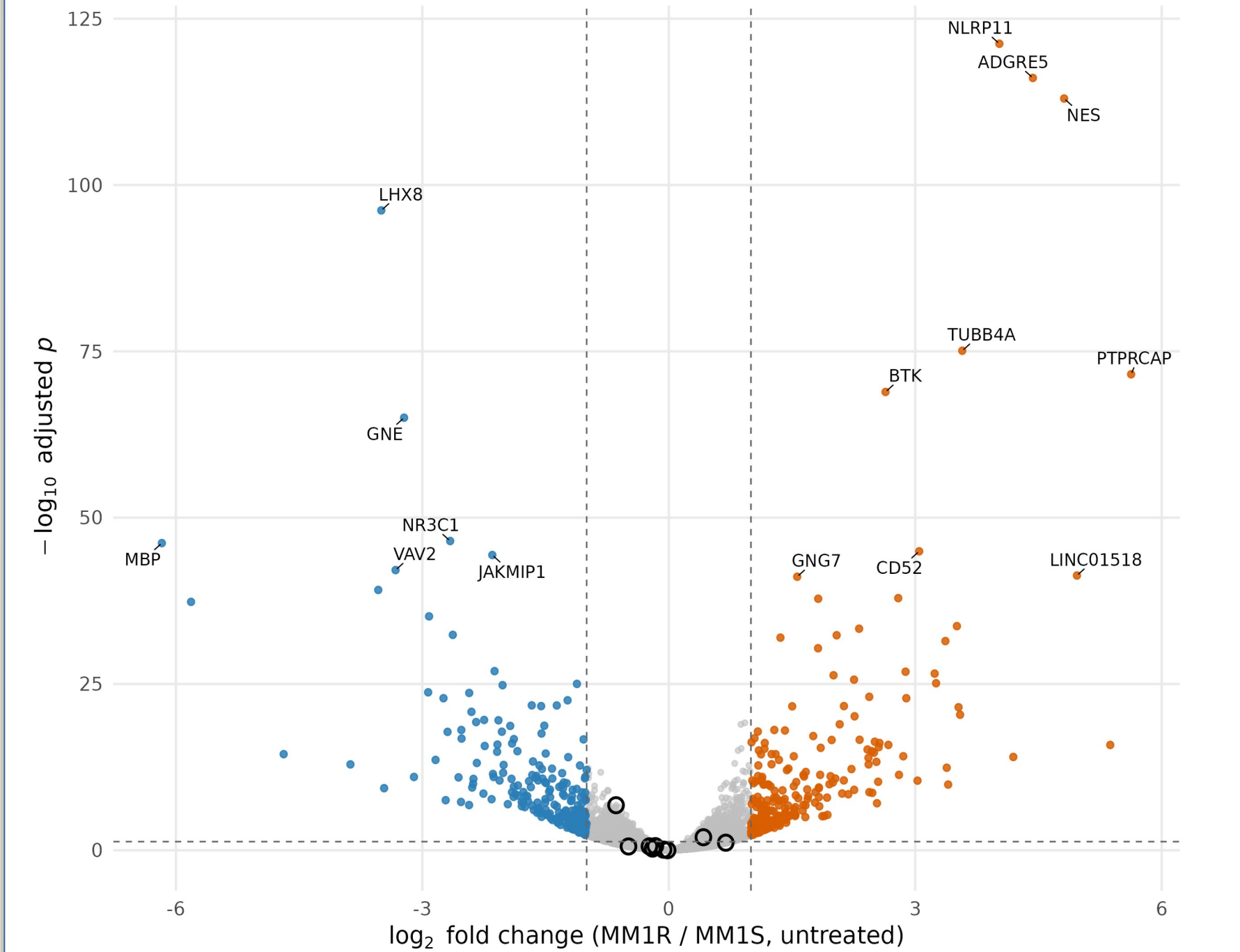
Ferroptosis-associated transcriptional landscape

GSE182638 : MM1R (Glucocorticoid-resistant) vs MM1S (Glucocorticoid-sensitive)



DIFFERENTIAL EXPRESSION BETWEEN MM1R & MM1S CELLS

DESeq2, n=3 vs n=3 (GSE182638) || circled points: A priori ferroptosis gene panel

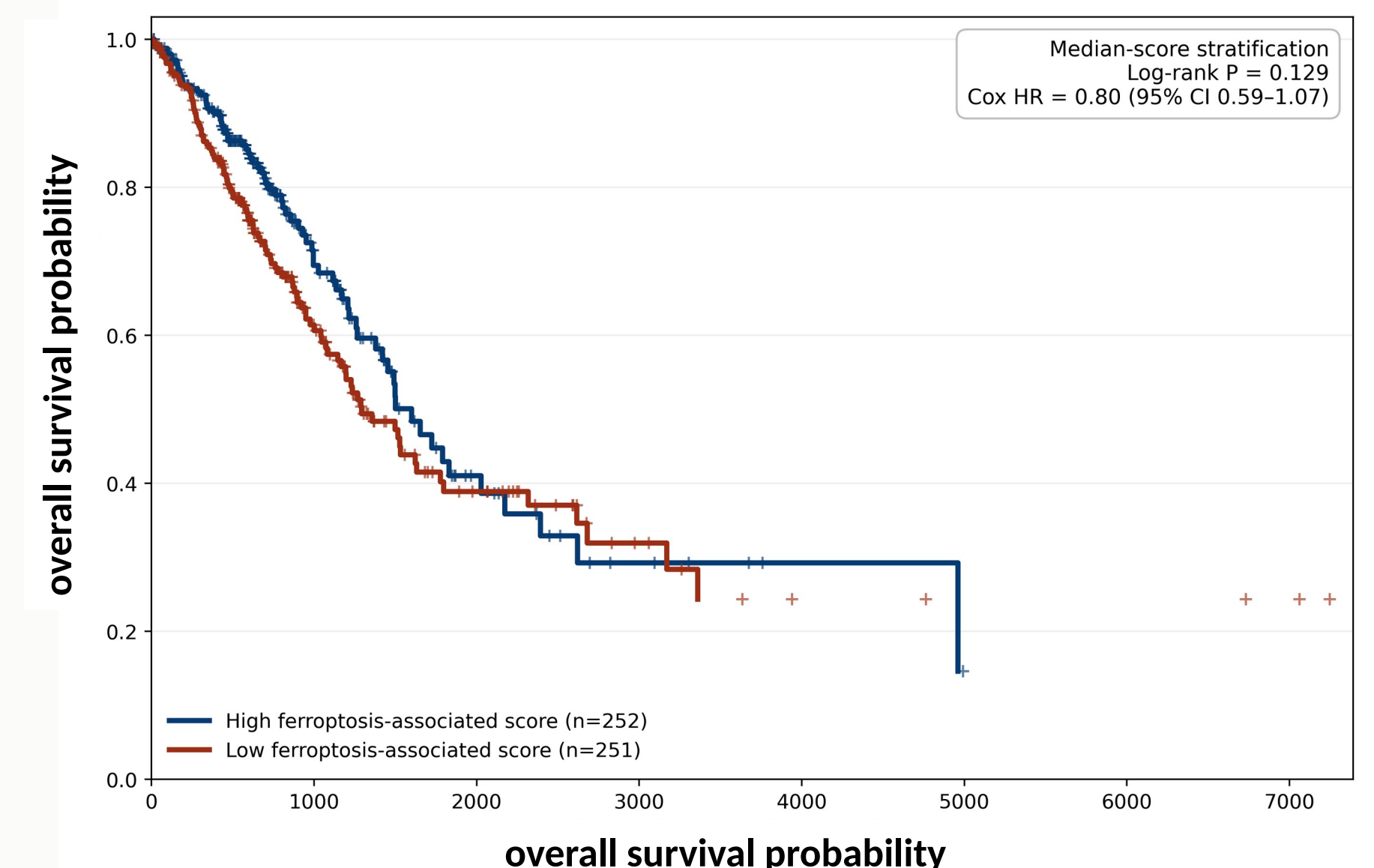


Hypothesis-generating: GSE182638 does not annotate ferroptosis sensitivity directly.

[Raw data were obtained from TCGA-LUAD (Cancer Genome Atlas Research Network, 2014) and processed using custom R scripts]

7. FERROPTOSIS SIGNATURE & SURVIVAL

Ferroptosis-associated molecular state and overall survival [TCGA-LUAD]



References :

1. Cancer Genome Atlas Research Network. Nature. 2014;511:543–550. doi:10.1038/nature13385.
2. Zhou N, Bao J. Database (Oxford). 2020;2020:baaa021. doi:10.1093/database/baaa021.
3. Greenstein S, et al. Exp Hematol. 2003;31:271–282. doi:10.1016/S0301-472X(03)00023-7.
4. Author-generated R analysis & visualizations; reproducible code: GitHub – github.com/humayrakabir-ahona/poster-presentation-data

